On Construction of Approximate Real Mutually Unbiased Bases for an infinite class of dimensions $d \not\equiv 0 \bmod 4$

Ajeet Kumar, Rakesh Kumar, Subhamoy Maitra, Uddipto Mandal

Indian Statistical Institute, Kolkata
Indian Institute of Technology, Kharagpur

8 June, 2026

Abstract

Motivation

- Real Mutually Unbiased Bases (MUBs) do not exist for any dimension $d > 2$ which is not divisible by 4.
- Thus, the next combinatorial question is how one can construct Approximate Real MUBs (ARMUBs) in dimensions which are not divisible by 4.
- In this paper, for the first time, we show that it is possible to construct $N(s) + 1$ number of ARMUBs where $N(s)$ is number of MOLS of order s for dimensions d of the form $d = (4n - t)s$, $t = 1, 2, 3$ where s and n are natural numbers. If s is an odd prime power, it is possible to construct $\geq \lceil \sqrt{d} \rceil$ many ARMUBs.
- Finally we connect this with Welch Bounds related to spherical codes.

Methodology

- Considering the objects from combinatorial designs, our method exploits any available $4n \times 4n$ real Hadamard matrix H_{4n} (if exists) and uses this to construct an orthogonal matrix Y_k of size $k \times k$, $k = (4n - t)$, such that the absolute value of each entry varies a little from $\frac{1}{\sqrt{k}}$.
- In our construction, the absolute value of the inner product between any pair of basis vectors from two different ARMUBs will be $\leq \frac{1}{\sqrt{d}}(1 + O(d^{-\frac{1}{4}})) < \frac{2}{\sqrt{d}}$, for proper choices of parameters, the class of dimensions d being infinitely large.

Mutually Unbiased Bases (MUBs)

Definition

Let

$$M_l = \{|\psi_1^l\rangle, \ldots, |\psi_d^l\rangle\}, \quad M_m = \{|\psi_1^m\rangle, \ldots, |\psi_d^m\rangle\}$$

be two orthonormal bases in $\mathbb{C}^d$.
These two bases are called Mutually Unbiased if

$$\left|\langle\psi_i^l|\psi_j^m\rangle\right| = \frac{1}{\sqrt{d}}, \quad \forall i, j \in \{1, 2 \ldots d\}.$$

Importance

- Quantum state tomography
- Quantum cryptography
- Quantum communication
- Signal processing and coding theory
- Combinatorics and finite geometries

MUBs correspond to maximally "uncorrelated" orthonormal bases.

Why Real MUBs are Difficult?

Known Facts

- In $\mathbb{C}^d$, maximum possible number of MUBs is $d+1$.
- Complete constructions are known only for prime power dimensions.
- Over $\mathbb{R}^d$, the situation becomes significantly more restrictive.

Key Obstruction

For dimensions

$$d > 2, \qquad d \not\equiv 0 \pmod 4,$$

there cannot exist even a pair of real MUBs.

Reason

Existence of a pair of real MUBs implies existence of a real Hadamard matrix. However, real Hadamard matrices exist only for orders divisible by 4.

Approximate Real MUBs (ARMUBs)

Approximate Mutual Unbiasedness

Instead of requiring exact equality $\left|\langle\psi_i^l|\psi_j^m\rangle\right| = \frac{1}{\sqrt{d}}$, we allow small deviations.

β-ARMUBs

A collection of orthonormal bases is called β-ARMUB if

$$\left|\langle\psi_i^l|\psi_j^m\rangle\right| \leq \frac{\beta}{\sqrt{d}},$$

for vectors belonging to different bases.

Goal

Construct many real orthonormal bases with $\beta \approx 1$.
Smaller β implies stronger approximation to ideal MUBs.

Existing Results on Real MUBs

Dimension d	Number of Real MUBs
d	$\leq d/2+1$
$4 \nmid d$	1
$4n$ with n not square	≤ 2 $= 2$ iff HM of size d exists
$4s^2$	≤ 3 ≥ 2 iff HM of size d exists
$4^i s^2$	$\geq MOLS(2^i s) + 2$ if HM of size $2^i s$ exists
4^i	$= d/2+1$

Reference: Boykin et al. 2018

ARMUBs: Existing Results and Our Contribution

Reference	Dimension	# Bases	Bound
Kumar et al 2021	$(4q)^2$	$\frac{\sqrt{d}}{4}+1$	$\frac{4}{\sqrt{d}}$
Kumar et al 2022	$q(q+1)$	$\lceil\sqrt{d}\rceil$	$< \frac{2}{\sqrt{d}}$
Kumar et al 2024	Several cases	Large	APMUBs
This Work	$(4n-t)s$	$\geq \lceil\sqrt{d}\rceil$	$\frac{1}{\sqrt{d}}(1+O(d^{-1/4}))$

Main Difference

Earlier constructions mainly required dimensions divisible by 4.
We construct ARMUBs for dimensions

$$d \not\equiv 0 \pmod 4.$$

Main Construction

Real Hadamard Matrix H_{4n}

$\Downarrow$

Construct ϵ-Hadamard Matrix

$\Downarrow$

Combine with Resolvable Block Designs

$\Downarrow$

Construct ARMUBs

ϵ-Hadamard Matrices

Definition

An orthogonal matrix $\mathbb{O}_{k\times k}$ is called an ϵ-Hadamard matrix if

$$|(\mathbb{O}_k)_{ij}| = \frac{1+\epsilon_{ij}}{\sqrt{k}},$$

where

$$\epsilon = \max\{\epsilon_{ij}\} = \pm O(k^{-\lambda}), \qquad \lambda > 0.$$

Interpretation

- Entries behave almost like Hadamard entries.
- Magnitudes remain close to
$$\frac{1}{\sqrt{k}}.$$
- Useful when exact Hadamard matrices do not exist.

This is the key new ingredient of our construction.

Block Matrix Reduction

Towards Constructing ϵ-Hadamard Matrices

We start from a real orthogonal matrix

$$H_{4n} = \frac{1}{\sqrt{4n}} \begin{bmatrix} U & V \\ W & D \end{bmatrix}.$$

$$\text{Here, } U : t \times t, \qquad D : (4n - t) \times (4n - t).$$

Using block matrix manipulations, we construct

$$Y_1 = \frac{1}{\sqrt{4n}} \left(D - \frac{1}{\sqrt{4n}} W \left(I + \frac{1}{\sqrt{4n}} U \right)^{-1} V \right), \text{ and}$$

$$Y_2 = \frac{1}{\sqrt{4n}} \left(D + \frac{1}{\sqrt{4n}} W \left(I - \frac{1}{\sqrt{4n}} U \right)^{-1} V \right).$$

Outcome

Y_1 and Y_2 become orthogonal matrices of order $(4n - t)$.

Main Reduction Theorem

Constructing ϵ-Hadamard Matrices

Theorem

Let $H_{4n} = \frac{1}{\sqrt{4n}} \begin{bmatrix} U & V \\ W & D \end{bmatrix}$ *be a real Hadamard matrix.*

If $t < \sqrt{n}$,

then the constructed matrices

$$Y_1, \; Y_2$$

are ϵ*-Hadamard matrices of order* $(4n - t)$, *with*

$$\epsilon = \frac{t}{\sqrt{k}} + O(k^{-1}).$$

Significance

We obtain ϵ-Hadamard matrices in dimensions where Hadamard matrices do not exist.

Explicit Cases: $t = 1, 2, 3$

Theoretical Result

Consider Y_1, Y_2, calculate two ϵ values accordingly, then consider the minimum one, and that will be upper bounded by the value as described in the table below.

t	Dimension	ϵ
1	$4n - 1$	$\frac{1}{2\sqrt{n}}$
2	$4n - 2$	$\frac{2}{\sqrt{n}}$
3	$4n - 3$	$\frac{4}{\sqrt{n}}$

Observation

The experimentally observed values of ϵ are often even smaller than the theoretical upper bounds.

Implementation

Python implementations and experiments are available in our repository.

Resolvable Block Designs (RBDs)

Basic Setup

Consider an RBD(X, A) with
$$|X| = d = ks.$$
Each block has size k.

Important Property of chosen RBDs

Any two blocks from different parallel classes should intersect in at most one point to achieve desired asymptotic values of β
$$\mu = 1.$$

How It Helps

- Each parallel class produces one orthonormal basis.
- ϵ-Hadamard matrices are used to embed vectors.
- Different parallel classes become approximately unbiased.

Main ARMUB Construction Theorem

Theorem

Let

$$d = ks,$$

where

$$s > k,$$

and s is an odd prime power.
If an ϵ-Hadamard matrix of order k exists, then one can construct

$$s \geq \lceil \sqrt{d} \rceil$$

many β-ARMUBs.

Main Bound

$$\beta \leq 1 + O(d^{-1/4}) < 2.$$

This gives infinitely many new dimensions with ARMUB constructions.

Why is this Significant?

Key Contributions

- No pair of real MUBs exists for

$$d > 2, \quad d \not\equiv 0 \pmod 4.$$

- We still construct many approximate real MUBs.
- Number of bases grows as

$$O(\sqrt{d}).$$

- Inner products asymptotically approach the optimal value

$$\frac{1}{\sqrt{d}}.$$

Conceptual Importance

This appears to be the first ARMUB construction framework for dimensions not divisible by 4.

Example

Consider

$$d = 79 \times 3^4.$$

Observation

- No pair of real MUBs exists in this dimension.
- Earlier constructions do not address this case.

Our Construction

- We obtain

$$3^4 = 81$$

  many ARMUBs.
- Since

$$79 = 4 \times 20 - 1,$$

  we obtain

$$\epsilon < \frac{1}{2\sqrt{20}}.$$

- Resulting parameter:

$$\beta < 1.252.$$

Connection to Spherical Codes

Geometric Interpretation

The vectors of ARMUBs may be viewed as points on a unit hypersphere.

Welch Bound

For N unit vectors in $\mathbb{R}^d$, if $N \leq d^2$

$$\tau_{\max} \geq \sqrt{\frac{N-d}{d(N-1)}}.$$

Our Setting

- Number of vectors:
$$N = O(d^{3/2}).$$
- Coherence approaches
$$\frac{1}{\sqrt{d}}.$$

Asymptotic Optimality

Main Observation

Our construction satisfies

$$\beta = 1 + O(d^{-1/4}).$$

Hence,

$$\beta \to 1 \qquad \text{as} \qquad d \to \infty.$$

Consequences

- Asymptotically approaches Welch Bound.
- Near-optimal spherical packing.
- Useful in:
  - coding theory,
  - compressed sensing,
  - CDMA,
  - vector quantization.

Conclusion

Summary

- Constructed ARMUBs for dimensions
$$d \not\equiv 0 \pmod 4.$$
- Introduced ϵ-Hadamard matrices.
- Obtained
$$\geq \lceil \sqrt{d} \rceil$$
many ARMUBs.
- Achieved
$$\beta = 1 + O(d^{-1/4}).$$
- Established connections with Welch Bounds and spherical codes.

Open Problems

Future Directions

- Can one construct APRMUBs for
$$d \not\equiv 0 \pmod 4?$$
- Can the bounds on ϵ be improved further?
- Is it possible to obtain asymptotically optimal constructions for all odd dimensions?
- Can computational search methods improve the parameters?

Any Questions?

Thank You